

Seongmin Hong

Postdoctoral Researcher, INMC, Seoul National University

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My research focuses on the **inversion of deep generative models**, spanning structural design and optimization-based methods. I develop principled algorithms for **exact and approximate inversion** of diffusion models and normalizing flows, with applications to **image editing**, **posterior sampling**, and **inverse problems in imaging**. My doctoral dissertation on this topic was recognized with the **Distinguished Ph.D. Dissertation Award**.

Research Keywords: Diffusion Model Inversion · Normalizing Flows · Latent Diffusion Models · Posterior Sampling · Inverse Problems · Optimization · Image Generation

Education

Ph.D. in Electrical and Computer Engineering – Seoul National University 03 2020 — 08 2025

Dissertation: Inversion of Deep Generative Models: Structural and Optimization-based Approaches

Award: **Distinguished Ph.D. Dissertation**

Advisor: Se Young Chun, GPA: 4.08/4.3

B.S. in Electrical and Computer Engineering – Seoul National University 03 2016 — 02 2020

GPA: 3.70/4.3

Math, Physics, and Astronomy – Gyeonggi Science High School for the Gifted 03 2013 — 02 2016

Experience

Postdoctoral Researcher (Alt. military service) – INMC, SNU 09 2025 — 08 2026

Honors & Fellowships

Apple Scholars in AI/ML PhD fellowship program - Institutional Nominee 2025

Qualcomm Innovation Fellowship Korea 2024 (won 4,000,000 KRW) 12 2024

Brain Korea 21 Four Program 03 2020 — 08 2025

Top-tier Conference Publications (First-authored)

[Gradient-free Decoder Inversion in Latent Diffusion Models](#)

Seongmin Hong, Suh Yoon Jeon, Kyeonghyun Lee, Ernest K. Ryu, Se Young Chun
Neural Information Processing Systems (**NeurIPS**), 2024.

[On Exact Inversion of DPM-Solvers](#)

Seongmin Hong, Kyeonghyun Lee, Suh Yoon Jeon, Hyewon Bae, Se Young Chun
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.

[On the Robustness of Normalizing Flows for Inverse Problems in Imaging](#)

Seongmin Hong, Inbum Park, Se Young Chun
International Conference on Computer Vision (**ICCV**), 2023.

[Neural Diffeomorphic Non-uniform B-spline Flows](#)

Seongmin Hong, Se Young Chun
Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI**, oral), 2023.

Other Conference Publications

Triadic Dynamics Aware Diffusion Posterior Sampling for Inverse Problems: Optimizing Guidance and Stochasticity Schedules
Junseo Bang*, Dong Ju Mun*, Hoigi Seo, **Seongmin Hong**, Se Young Chun
International Conference on Machine Learning (ICML), 2026.

Preprints & Manuscripts under Review

[Principled Reframing in Motion Transfer with Video Diffusion Models](#)
Donggyu Lee*, **Seongmin Hong***, Se Young Chun (*co-first authored), 2025

Invited Talks

[Inversion of Deep Generative Models: Structural and Optimization-based Approaches](#)
At Chung-Ang University, Seoul, Korea, Oct 22, 2025

[Inversion of Deep Generative Models: Structural and Optimization-based Approaches](#)
At ETRI, Daejeon, Korea, Oct 21, 2025

[On the Robustness of Normalizing Flows for Inverse Problems in Imaging](#)
In 2024 SIAM Conference on Imaging Science (**SIAM IS24**), Atlanta, Georgia, U.S., May 28-31, 2024

Teaching Experience

Head TA, Topics in Signal Processing (Generative Diffusion Model, English Lecture)	2025 Autumn
Head TA, Signals and Systems (English Lecture)	2022 Spring
Head TA, Seminar in Electrical and Computer Engineering 3	2020 Spring
TA, Advanced Electromagnetics 1	2020 Spring

Mentoring & Advising

Research Mentor for 10+ Undergraduate Interns <i>Intelligent Computational Imaging Lab (ICL), SNU</i>	2022 — Present
•Inbum Park (Now Ph. D. student at University of Maryland) — <i>ICCV 2023</i> •Kyeonghyun Lee (Now Ph.D. student at Seoul National University) — <i>CVPR 2024</i> •Donggyu Lee (Undergraduate Intern) — <i>Preprint 2025</i> •Mentored 3+ additional interns on their early-stage research and career development.	

Academic Services

Technical Research Personnel (Alternative military service)	03 2023 — 02 2025, 09 2025 — 08 2026
Reviewer: NeurIPS (2024, 2025), ICML (2025, Gold 2026), ICLR (2025, 2026), CVPR (2025), ICCV (2025), AAAI (2026), AISTATS (2025, 2026), ACMMM (2025) IEEE Sensors Journal, ACM TOMM, IEEE TPAMI, IEEE TMM	since 2023
· ICML 2026 Gold Reviewer: awarded to top 25% of reviewers based on area chair ratings	

References

Prof. Se Young Chun (Advisor)
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Prof. Ernest K. Ryu
Assistant Professor, Applied Mathematics
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